RESEARCH PROPOSAL

Digital Twins for Cyber-Resilient Power Grids

A systematic review and research agenda for the AI-powered era

100
papers screened and included

2020–26
coverage window

83
full texts retrieved

5.1%
engage AI-specific threats

Prepared from a PRISMA-screened corpus · October 2026

Motivation: why this, why now

The grid is becoming software-defined — and software is attackable

- **Digitisation is accelerating.** Smart meters, SCADA/DCS, distributed energy resources, EV charging and 5G/6G links widen the cyber-physical attack surface of the electricity grid.
- **Digital twins are being deployed at scale.** Grid operators use twins for monitoring, predictive control, resilience and operator training — the market is forecast to grow several-fold this decade.
- **Twins are marketed as a security tool.** They enable safe attack rehearsal and twin-vs-plant divergence detection.
- **Offensive AI is maturing.** Adaptive evasion, data poisoning, and LLM/agentic attack automation now target exactly the ML that modern defences rely on.
- **The open question.** Is grid-digital-twin defence keeping pace — or is it adopting AI faster than it models AI as the threat?

TENSION

The digital twin is simultaneously the defence's best sensor and a newly exposed attack surface — yet the two roles are almost never studied together.

Problem statement and gap

Defence has adopted AI faster than it has modelled AI — or the twin — as the target

- **Defensive bias.** The field builds detectors; far less work studies how those detectors and the twin itself can be attacked.
- **Circular dependency.** Twins are trusted to catch grid attacks while nothing checks attacks on the twin.
- **Unmodelled adversary.** Threat models remain classical (FDI, replay, DoS) — not adaptive or AI-enabled.

EVIDENCE SNAPSHOT

Only 5.1% of relevant works (24 of 469) engage any AI/ML-specific attack vector. In the top-100 corpus: adversarial ML 0 · evasion 0 · poisoning 0.

Toward a research question

- How do adversarial and defensive studies of grid digital twins compare in scope, threat model and evaluation?
- And to what extent does the literature confront AI-enabled threats against the twins themselves?

Research questions

Four questions structure the review and the agenda

RQ1

Landscape

What is the state of research on digital twins for power-grid cybersecurity — methods, threat models, venues and maturity?

RQ2

Asymmetry

How do adversarial (attack-focused) and defensive (detection/mitigation) studies differ in object of study, assumed adversary and evaluation?

RQ3

AI-era gap

To what extent does the literature address AI-enabled threats (evasion, poisoning, LLM/agentic attacks) against grid digital twins?

RQ4

Agenda

What research priorities and safeguards would make grid digital twins robust in an AI-powered threat environment?

Objectives and scope

OBJECTIVES

- Build a reproducible, PRISMA-screened corpus of digital-twin grid-cybersecurity research.
- Classify every study by stance: adversarial, defensive, bridging, review or enabling.
- Compare adversarial and defensive strands on a common analytical framework.
- Quantify engagement with AI-specific threats and identify blind spots.
- Derive a prioritised research agenda and design implications for operators.

SCOPE

- Window: 2020 – 2026, with in-press 2026 records retained.
- Included: digital twin + power/grid + cyber/security.
- Excluded: IT-only digital twins, non-grid energy, purely physical twins.
- Sectors noted beyond the grid: nuclear (adversarial robustness), EV, telecom microgrids.
- Units: peer-reviewed articles, conference papers, theses and technical reports.

Background: the digital twin and the grid threat landscape

Digital twin anatomy

- A living model: physical asset + real-time data + bidirectional interaction.
- Grid twins span substations, DERs, protection relays, EV charging and markets.
- Deployed for monitoring, predictive control, resilience and education.

Grid cyber-physical risk

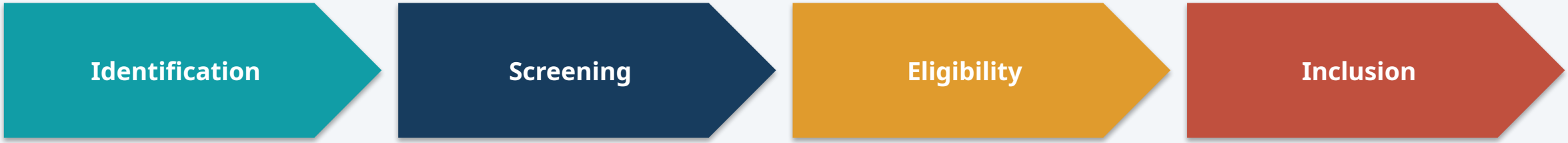
- Attack surface: SCADA/DCS, AMI/meters, DER interfaces, comms and market data.
- Known threats: False Data Injection, replay, DoS, supply-chain, APTs, hardware trojans.
- Consequence: physical — outages, equipment damage, market manipulation.

Prior evidence base

- Electric-DT grid frameworks (2022) and IoT-DT microgrid resilience (2020) are heavily cited.
- Recent surveys emphasise AI, blockchain and co-simulation.
- Gap confirmed: surveys catalogue threats but rarely test author-labelled defences.

Methodology: a PRISMA-style systematic workflow

Reported to PRISMA 2020 conventions; every stage reproducible



1,287 unique works

12 core OpenAlex queries + adversarial-AI probes

492 eligible

title/abstract on DT + grid + security

100 retained

ranked by relevance, citations, recency

100 papers / 83 PDFs

classified, venue-mapped, verified

SOURCES, VERIFICATION AND TOOLS

- Discovery: OpenAlex (+ Unpaywall, Semantic Scholar); literacy verification against Crossref and DataCite.
- Full-text retrieval: open-access PDFs harvested programmatically; browser-assisted recovery where publishers block automation.
- Analysis: automated stance tagging with manual adjudication; abstract- and full-text keyword profiling for AI-threat engagement.

Corpus: from 1,287 records to 100 included studies

Screening funnel with every exclusion accounted for

1,287

unique works identified

from 12 OpenAlex queries

492

eligible after screening

795 excluded: no twin term or no grid domain

100

included in final corpus

relevance + citation + recency ranking

83

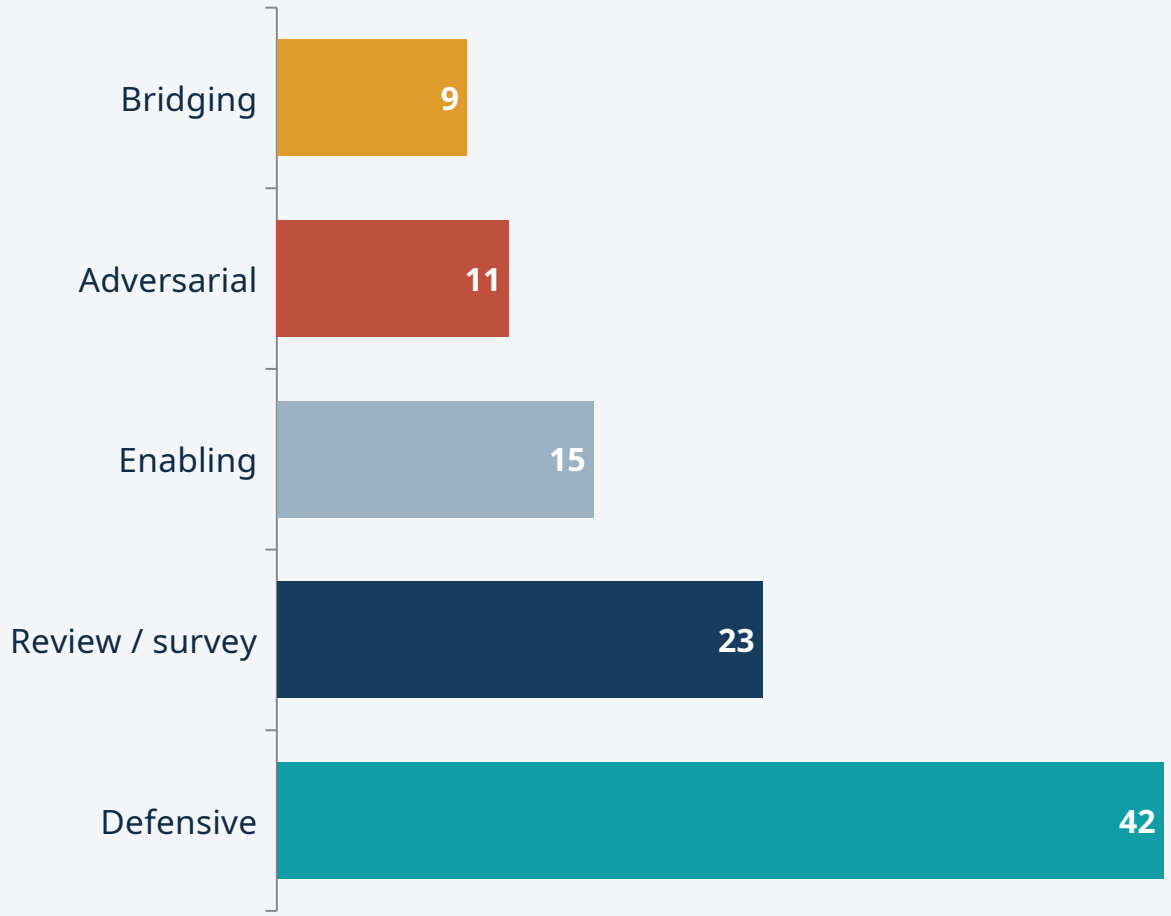
full texts retrieved

open access, validated PDFs on disk

INCLUSION AND QUALITY CONTROLS

- Every DOI verified against Crossref/DataCite with byte-exact title matching.
- Two abstracts-flagged 2026 in-press records retained and re-verified.
- Venue labels corrected where OpenAlex metadata disagreed with the registry.
- Anomalies logged: 3 DOI-less repository items, 1 non-resolving DOI.

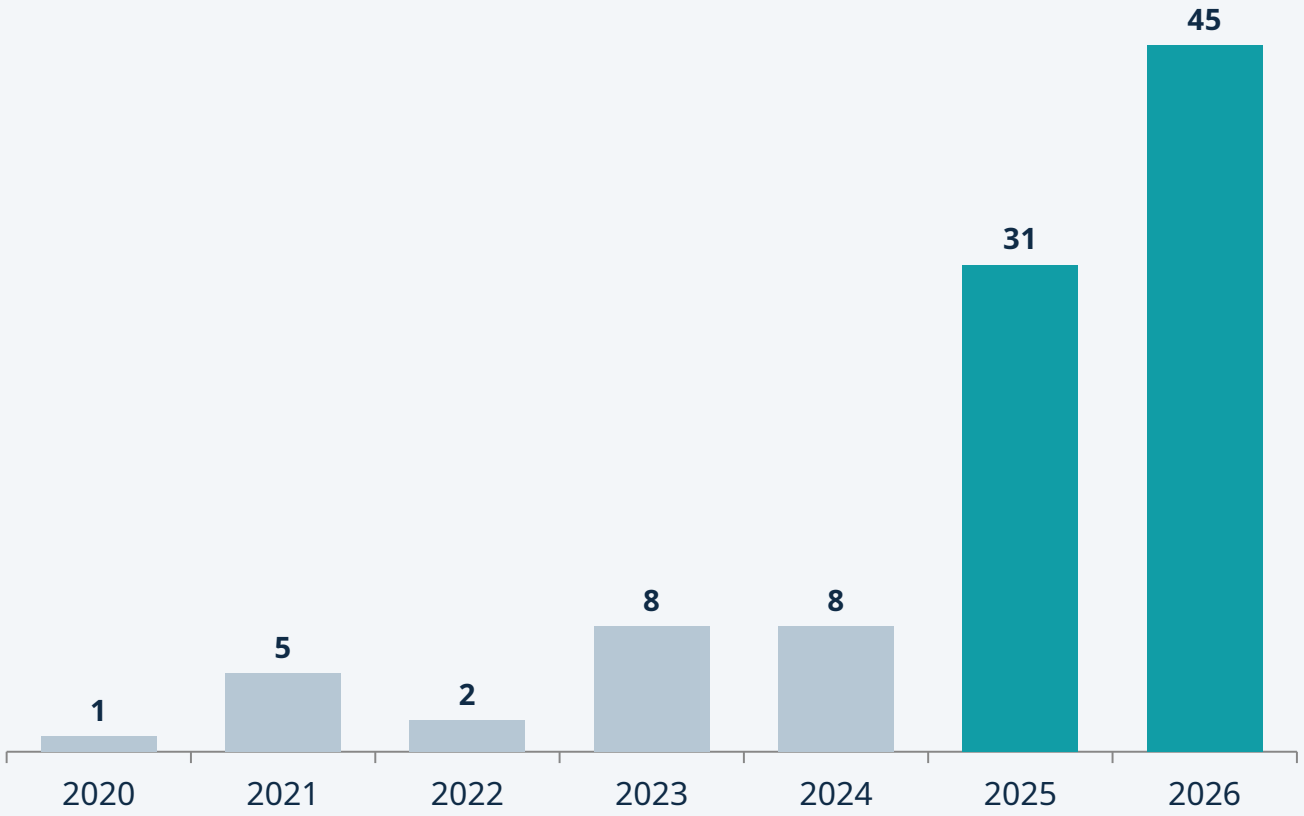
Corpus composition: the field is overwhelmingly defensive



READING THE DISTRIBUTION

- **44 of 100 act on security.** 42 defensive plus 2 defensive-leaning bridging studies build or test protection.
- **Only 11 are adversarial.** 1 in 4 security studies even asks how the twin could be broken.
- **9 bridge the two.** They pair a threat model with a detector or co-simulation testbed.
- **38 report rather than act.** Reviews and enabling studies catalogue the field.

Corpus: venues, access and recency



WHAT THE SHAPE TELLS US

- **76% published in 2025–26.** The field is young and accelerating; 2026 alone holds 45 papers.
- **Venues are fragmented.** 100 papers across ~79 venues — no single dominant outlet.
- **Access is uneven.** 83 full texts retrieved; 17 remain paywalled or bot-blocked.
- **Implication.** Recency plus fragmentation makes a systematic map genuinely useful.

Preliminary findings: adversarial vs defensive research

Dimension	Adversarial papers (11)	Defensive papers (42)
Research question	Can it be broken, and how?	Can we catch or absorb it?
Object of study	The twin interface, threat model, attack chain	The detector, controller, recovery loop
Role of the twin	Target — its interface is the attack surface	Sensor — twin-vs-plant divergence
Adversary assumed	Creative but classical (FDI, replay, DoS, APT)	The same fixed catalogue of attacks
Evaluation	Attack demonstrates a breach	Detection accuracy vs a predefined attack set
Hidden assumption	Attackers adapt	The twin's data and model are trustworthy

The two literatures barely share a threat model: one treats the twin as the thing to compromise, the other as the sensor that cannot lie.

Preliminary findings: the AI-as-attacker blind spot

0
works on adversarial ML or adversarial examples

0
works on evasion of ML-based detectors

0
works on data/model poisoning or backdoors

1
work attacking the twin's ML model (nuclear sector)

WHAT THE CORPUS DOES AND DOES NOT COVER

- **5.1% engage AI-specific threats.** Just 24 of 469 relevant works touch any adversarial-ML, poisoning or LLM vector.
- **LLM-in-the-loop twins: 1 paper.** Prompt injection and model extraction: absent.
- **Agentic adversaries: 2 papers.** Autonomous/agentic attack automation is essentially unstudied for grids.
- **Adversarial robustness exists — elsewhere.** The strongest attack-on-model work is in nuclear digital twins, not electricity.

Expected findings

Hypotheses the completed review will test and quantify

H1 — **Structural defensive bias**

Security research on grid twins will remain ~4:1 defensive over adversarial, with attack studies concentrated in threat modelling rather than adversarial robustness.

H2 — **Minimal AI-threat engagement**

Fewer than ~10% of relevant studies will substantively model AI-enabled attacks; evasion and poisoning will be near-absent.

H3 — **Trust boundary is unexamined**

Detector-based defences will overwhelmingly assume trustworthy twin data and models, leaving the twin's own ML as an unaudited attack surface.

H4 — **Imported practice, not home-grown**

Adversarial-robustness methods will appear first in adjacent critical sectors (nuclear, vehicles) and only belatedly transfer to electricity grids.

Contributions and significance

A systematic map	The first PRISMA-reported review of digital twins specifically for power-grid cybersecurity, spanning 2020–2026.
A reproducible corpus	100 screened records with verified DOIs, venue mapping and 83 retrieved full texts — an auditable research artefact.
An adversarial–defensive framework	A common analytical lens comparing threat models, evaluation practice and hidden assumptions across strands.
A quantified gap	The first measurement of how little grid-twin research engages AI-specific threats.
A research agenda	Prioritised directions: adversarial-robust detectors, poisoning-resistant twins, and LLM/agent threat models for AI-in-the-loop twins.

Roadmap, limitations and next steps

WORK PLAN

Months 1–2	Finalise corpus and screening log; publish reproducibility package.
Months 3–5	Full-text extraction: threat model, method, evaluation, assumptions.
Months 6–7	Adversarial–defensive comparison and AI-threat coverage scoring.
Months 8–9	Gap model: map unaddressed attack vectors to twin components.
Months 10–12	Research agenda, expert validation, write-up and submission.

LIMITATIONS

- Metadata sourced from OpenAlex; venue labels occasionally disagree with registries.
- Abstract-truncated profiling can under-count AI-threat terms; full-text pass will correct this.
- Coverage is English-language and overwhelmingly journal-indexed.
- Very recent in-press 2026 work may be under-represented.

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The 100 included studies

DOIs verified against Crossref and DataCite · 83 full texts retrieved

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DOIs of the 100 included studies — page 1 of 3

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